

SEGUN FADIPE

Backend Developer & Automation Engineer

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[GitHub](#)

Lagos, Nigeria

OBJECTIVE SUMMARY

Backend Developer with 2+ years of experience building event-driven systems, job queue architectures, and automated backend workflows. Focused on improving API performance, reducing response times, and designing scalable systems using caching and background processing. Recently expanding into AI-powered automation and LLM-integrated systems.

TECHNICAL SKILLS

Backend:

Node.js, Express.js, REST API Design,
Nest.js (basic knowledge)

BullMQ, Node-cron, Job Queue Architecture

Database & Caching:

MongoDB, Mongoose, Schema Design,
Aggregation Pipelines

Redis, Caching Strategies, TTL Indexes

Automation & Workflow:

n8n, Zapier (basic knowledge), Prompt
Engineering, Webhooks and API integration, AI
integration

Frontend (Supporting):

EJS, HTML5, CSS3, Tailwind CSS

React.js, Next.js

Tools & DevOps:

Git, GitHub, Postman

PM2, Oracle Cloud, Railway

Docker (basic)

Language:

JavaScript

PROJECT EXPERIENCE

1. Curio – Personalized News Digest Platform

Tech Stack: Node.js, Express.js, MongoDB, Redis, BullMQ, N8n, Nodemailer, EJS, Tailwind CSS

GitHub: <https://github.com/SegunNano/curio>

Live: <https://curio-news-app.up.railway.app/>

- A backend-heavy news aggregation and digest delivery system that fetches, caches and delivers personalized daily news to users via automated email workflows.
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- Architected a queue-based system using BullMQ and Redis handling async email delivery, news fetching and inactivity detection across multiple workers
- Implemented category-level Redis caching strategy shared across all users, significantly reducing redundant MongoDB queries
- Built MongoDB aggregation pipeline grouping and slicing articles by category at DB level, eliminating in-memory filtering
- Designed concurrent email processing using Promise.allSettled and p-limit, controlling throughput to 10 concurrent sends
- Integrated NewsData.io API fetching 130+ articles daily across 13 categories with TTL-based auto-deletion after 7 days
- Implemented intelligent digest scheduling – querying only due users directly in MongoDB using \$expr operators, reducing application-level filtering
- Built automated inactivity detection marking users inactive after 30 days using Redis pipeline for bulk cache invalidation
- Applied security best practices: helmet, rate limiting, mongo sanitization, HPP, JWT httpOnly cookies
- Refactored codebase applying DRY principles – centralized constants, shared queue configs, reusable utility functions

2. URL Shortener

Tech Stack: Node.js, Express.js, MongoDB, EJS, Tailwind CSS, Passport.js

GitHub: https://github.com/SegunNano/url_shortener

Live: <https://nanourl-0s58.onrender.com/>

- Built URL shortening service with custom short codes and automatic redirection
- Implemented hash generation for unique short links with MongoDB for fast lookups
- Integrated user authentication with email verification and automated password reset
- Used Node-cron to reset monthly link quotas ensuring fair usage

3. Ludo King – Multiplayer Game

Tech Stack: Node.js, Express.js, Socket.IO, EJS, Tailwind CSS, Passport.js

GitHub: <https://github.com/SegunNano/ludo-king>

- Built real-time multiplayer Ludo game with online room creation and joining
- Used Socket.IO for synchronized dice rolls, player turns and game state management
- Implemented complete game logic including turn management, rule enforcement and win conditions
- Optimized for low latency to ensure smooth real-time gameplay

4. MERN E-Commerce Platform

Tech Stack: MongoDB, Express.js, React.js, Node.js, Tailwind CSS, Firebase

GitHub: <https://github.com/SegunNano/store>

- Built full-stack e-commerce platform with custom RESTful APIs
- Implemented user authentication, admin panel and payment gateway integration
- Integrated Firebase for image uploads and real-time notifications
- Secured with hashed passwords and JWT-based authentication

EDUCATION

BSc Botany 2024

University of Lagos, Akoka, Lagos state,

LANGUAGE

English: Native Language, Distinguished levels in Listening, Speaking, Reading and Writing.

REFERENCES

Available upon request.